

African Cassava Mosaic Disease (ACMD)

Cassava Mosaic Virus Khate la chinangwa

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Symptoms on the leaves caused by Cassava Mosaic Virus (Eric Boa)  White fly infestation on the underside of a cassava leaf (Neil Palmer, CIAT)  Adult white fly, <i>Bemisia tabaci</i> (©John Innes Institute)	- This viral disease is spread mainly through insected cuttings used as planting material and is also transmitted by whiteflies (<i>Bemisia tabaci</i>). - Planting material can look healthy but the disease can be present, so use plant certified disease-free cuttings from registered suppliers. Use resistant varieties: Sauti, Sagonja, komanso, Beatrice, Mkondezi, Maunjili, Silira, Yizaso, Phoso, Mulola, Mpale, CH06/033, CH05/143. - Avoid mixing planting materials i.e. certified and non-certified planting materials. - Sterilize tools used to prepare cuttings with fire or disinfectant such as sodium hypochlorite, 70% ethanol (Methylated spirit). - Practice crop rotation with maize or sweet potato every other growing season. - Remove weeds to discourage habitat for the vectors. - Intercrop with maize or cowpeas to repel whiteflies. - Avoid disease hotspot areas for seed production.	- Check the cassava plants weekly for symptoms as soon as the plants start to sprout. - Look for yellow/white mottled spots on the leaves; wrinkled and deformed leaves; and the presence of small whiteflies on the leaves especially on the lower side of the leaf. - Also look for yellowing and very tiny leaves, tiny stems, stunted growth and small tubers. - Apply green direct controls as soon as disease symptoms are seen and yellow direct controls when whiteflies are present.	- Uproot all infected plants once they are observed from the field/garden. Dry and burn far from the field. - Create a conducive environment for natural predators such as lacewings, mites of the genera <i>Amblyseius</i> spp and <i>Typhlodromus</i> spp and aphelinid parasitoids such as <i>Encarsia formosa</i> and <i>Eretmocerus</i> spp, minute bugs and <i>Delphastus pusillus</i>.	- Use personal protective equipment when handling and applying pesticides, follow the label recommendations. Do not eat, drink or smoke when mixing or applying pesticides. - Do not apply insecticides in areas where cassava leaves are harvested for food. - Judicious use of pesticides to preserve natural enemies. Using insecticides to manage insect vectors of ACMD is not advised for small-scale farmers, it is usually ineffective and is not cost effective. - Apply cypermethrin 20g/L (Cypermethrin 20EC; Cyrux 20EC) at a rate of 5ml per 20L water. Contact and stomach acting pyrethroid. IRAC group: 3A - Apply deltamethrin 25g/L (Decis EC; Deltamethrin EC) at a rate of 70ml per 100L water. Pyrethroid with contact and residual activity. IRAC group: 3A	- For larger producers growing planting materials (cuttings) it might be advisable to use insecticide to manage white flies using the prescribed insecticides - WHO II - Moderately hazardous. Apply when whiteflies are noticed. Very toxic to aquatic animals, do not spray near water. - WHO II - Moderately hazardous. Apply when whiteflies are noticed. Very toxic to aquatic animals, do not spray near water.



Malawi

CREATED/UPDATED: June, 2016/December 2022

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